

Global Snow Persistence Explorer

Project Overview

Global Snow Persistence Explorer is an interactive Google Earth Engine application for exploring annual snow persistence worldwide using ERA5-Land data. Daily snow cover data are aggregated by calendar year to calculate snow-covered days and annual snow persistence, which is then classified into four snow persistence zones using thresholds adapted from Saavedra et al. (2017).

The application maps these zones, calculates their global area, and provides a location inspector for examining snow-covered days, annual snow persistence, monthly mean snow water equivalent (SWE), and glacier or ice cap information. The application supports complete calendar years from 1951 to 2025.

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